

KARIGAR

Artisans, Amplified

AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans

PROJECT REPORT

Problem Statement ID	SIH26090
Theme	Culture & Heritage
Category	Software
Team	Team Karigar (Team ID 16)

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Sections 8 and 9 are the team's proposed planning content, added to complete this report. All other sections are drawn from the Team Karigar presentation.

1. Executive Summary

India is home to an estimated 64.66 lakh handloom and handicraft artisans, many of whom work in rural settings with limited formal education and limited access to digital tools. Their products carry deep cultural value, yet their reach is often confined to local markets, seasonal exhibitions and intermediaries who capture a large share of the price.

Karigar is a mobile-first, AI-driven platform that lets artisans list, price and sell their work using only a phone camera and their own voice, in their own language. The application converts raw product photographs and spoken descriptions into professional catalogue entries, recommends fair prices, connects artisans with B2C, B2B and institutional buyers, supports custom orders through direct artisan-to-customer communication, and attaches verifiable provenance to every product.

The solution is designed to run on a basic 2 GB RAM smartphone, uses a proven technology stack (TypeScript, Node.js, PostgreSQL, with Python for custom AI models where needed), and follows a low-commission model that aligns platform success with artisan income.

64.66 lakh

handloom and handicraft artisans in India (PIB, Ministry of Textiles, 2025)

23.2%

of rural handloom workers had never attended school (Fourth All India Handloom Census, 2019-20)

71%

of weavers are women (Ministry of Textiles data, Aug 2025)

2. Problem Statement and Context

The problem statement (SIH26090) calls for an AI-driven market linkage and smart cataloging mobile application for marginalized artisans, under the theme Culture and Heritage. The team identified four core barriers that prevent artisans from participating fully in the digital economy.

Barrier	Description
Discoverability	Artisans struggle to reach customers beyond local markets and intermediaries. Sales depend heavily on exhibitions and fairs, such as Ghe Bharari in Maharashtra.
Digital barriers	Creating product listings, inventory records, descriptions, pricing and digital catalogues is difficult for first-time digital users, especially those with limited literacy.
Custom orders	Conventional e-commerce is built around ready-made products and does not support direct artisan-customer customisation of design, size, colour or quantity.
Authenticity	Customers cannot easily verify who made a product, where it came from, or its craft heritage, which weakens trust and undermines fair pricing.

Existing government initiatives demonstrate strong policy support for the sector. The One District One Product (ODOP) scheme spans 1,243+ products across 775 districts, and the Government e-Marketplace (GeM) ODOP Bazaar lists handicrafts across 446+ categories. Karigar complements these efforts by addressing the last-mile challenge: enabling the individual artisan to create, manage and sell listings without technical or linguistic barriers.

3. Proposed Solution

Karigar is organised as a seven-stage journey that takes an artisan from registration to growth analytics, wrapped in a secure and trusted platform layer.

Stage	Module	Key capabilities
1	Artisan Onboarding	Simple mobile or CSC-assisted sign-up; profile creation and KYC; bank and payment setup; multi-language support.
2	Smart Cataloging (AI-powered)	Artisan uploads raw photos; AI generates title, category, description, tags and craft style or origin; suggests a price range; produces SEO-friendly content.
3	Product Inventory and Listing	Inventory management; variants such as size and colour; stock updates and bulk upload; preview and publish to marketplace.
4	Dynamic Pricing Engine	AI-based price recommendation using market demand analysis, material cost, competitor analysis, seasonality and sales history.
5	Market Linkage Hub	Reach B2C customers (pan-India and global); B2B buyers (retailers, wholesalers, boutiques, exporters); government e-marketplaces (GeM, state emporiums, handicraft boards); digital participation in exhibitions and craft fairs.
6	Customizations and Orders	Custom orders and requirements; in-app chat with the artisan; confirmation of design, size, colour and quantity; secure order management.
7	Analytics and Growth Tools	Sales dashboard and reports; top products and customer insights; revenue analytics; promotions, discounts and social sharing.

Trust layer

Across all stages, the platform provides an AI authenticity assistant, provenance and craft-story records, secure payments with escrow, support and training, and a multi-language interface. Together these turn each product listing into a verifiable record of maker, origin and heritage.

4. Technical Approach

The architecture favours mature, widely supported technologies so that the platform can be built, maintained and scaled by a typical development team, while AI capabilities are delivered through a combination of hosted services and custom models.

4.1 Technology stack

Layer	Technology	Purpose
Frontend	TypeScript / TSX	Mobile-first user interface, lightweight enough for low-end devices.
Backend	Node.js with TypeScript	APIs, business logic, order and inventory services.
Database	PostgreSQL with SQL	Relational storage for artisans, products, orders and provenance records.
AI / ML	Python (where custom models are needed)	Custom models for matching, pricing and anomaly detection.
Storage	Cloud storage	Secure, backed-up image storage with retrieval from any device.

4.2 AI components

Component	Function
AI Image Enhancement	Background removal, lighting correction and quality enhancement, so a phone photograph becomes a marketplace-ready image.
Speech-to-Text (Whisper)	Converts the artisan's spoken description into text in regional languages.
Large Language Model (GPT / Gemini)	Turns transcribed speech into a structured product catalogue entry and a compelling description.
Translation AI	Translates Marathi, Hindi and other Indian languages into the customer's language, and back.
AI Pricing Assistant	Recommends a fair price or price range from cost, demand and competitor signals.
AI Market-Matching Engine	Matches each artisan with suitable consumer and B2B buyers.
Fraud / Anomaly Detection AI	Flags suspicious listings, orders and user behaviour to protect artisans and buyers.

4.3 Voice-to-listing workflow

The artisan photographs the product and describes it aloud in their own language. Speech-to-text transcribes the description, the language model structures it into catalogue fields, translation prepares versions for customers, the image pipeline enhances the photograph, and the pricing assistant proposes a range. The artisan reviews and confirms the listing with a single tap before it is published.

Device accessibility: the application is designed to be usable on a basic 2 GB RAM mobile phone, with heavy AI processing performed in the cloud rather than on the device.

5. Feasibility Analysis

Dimension	Assessment
Technical	AI cataloging, image recognition and cloud-based delivery are established technologies with commercially available services, reducing development risk.
User-friendly	Photo-based and voice-based listing removes the need to type, which is the main barrier for first-time digital users.
Multilingual	Regional languages, Hindi and English are supported through AI translation, allowing artisans and customers to communicate in their preferred language.
Scalable	Cloud infrastructure and a standardised onboarding flow allow more artisans and products to be added without redesign.

6. Viability Analysis

Factor	Description
Low platform fee	A small commission on successful sales only, so artisans pay nothing unless they earn.
Wider reach	Access to customers beyond local markets and districts.
Year-round sales	Products remain available throughout the year rather than only during fairs or seasons.
More customers	Access to B2C, B2B and institutional buyers.
Social impact	Higher income, fewer middlemen and preservation of craft traditions.

The team's projection is that the share of artisan sales made through digital applications will rise over time while the share made through exhibitions and fairs declines. This is an illustrative projection and should be validated through pilot data.

7. Impact and Benefits

Impact

- Direct market access for artisans, with no middlemen and better price realisation.
- Wider visibility and preservation of India's regional craft traditions.
- Verified, personalised products that build customer trust in authentic Indian art.

Benefits by stakeholder

Stakeholder	Benefit
Economic	Higher, fairer margins for artisan sellers.
Social	Empowers low-literacy, rural sellers, including the women who make up most of the weaver workforce, to run their own business.
Cultural	Documents and promotes heritage crafts through provenance history.
Consumer	Personalised, authenticity-verified purchases with optional offline discovery.

8. Risks and Mitigation

Proposed planning content, not part of the original presentation.

Risk	Mitigation
Low digital literacy limits adoption	Voice-first design, assisted onboarding through CSC centres, and in-app training.
Poor connectivity in rural areas	Lightweight app, offline draft saving and background sync when a connection is available.
Inaccurate AI descriptions or translations	Artisan confirmation step before publishing; feedback loop to improve models.
Fake or misattributed products	Provenance records, KYC, and fraud and anomaly detection.
Payment disputes or non-delivery	Escrow-based payments and secure order management.
Data privacy of artisan information	Consent-based data use, encrypted storage and access controls.

9. Proposed Implementation Roadmap

Proposed planning content, not part of the original presentation. Timelines are indicative.

Phase	Focus	Outcome
Phase 1: Prototype	Onboarding, photo and voice-to-catalogue flow, basic listing	Working demo in Hindi, Marathi and English
Phase 2: Pilot	Onboard artisans in selected districts with partner support; add pricing assistant and orders	Real-user feedback and measured time-to-list
Phase 3: Marketplace	Market linkage hub, custom orders, escrow payments, authenticity and provenance	Buyers across B2C, B2B and institutional channels
Phase 4: Scale	Analytics, additional languages, government marketplace integration	Multi-state rollout

Suggested success indicators

- Number of artisans onboarded and active listings created.
- Average time taken to publish a listing.
- Increase in artisan income and share of sales outside the local market.
- Repeat-buyer rate and custom-order completion rate.

10. Conclusion

Karigar addresses the four barriers that hold back India's artisans: discoverability, digital skills, customisation and authenticity. By combining voice-and-photo-based AI cataloging with fair pricing, market linkage and verifiable provenance, it reduces the effort of going digital to a single conversation with a phone. The solution is technically feasible with mature technologies, viable through a low-commission model, and aligned with national efforts such as ODOP and GeM. Above all, it advances the goal of the Culture and Heritage theme: empowering artisans, expanding markets and preserving heritage.

References

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